

Optimizing e-commerce rankings: insights and implications for Amazon sellers

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Outline

- ① Literature Review
- ② Research gap and questions
- ③ Ranking model (RQ1)
- ④ Survival analysis model (RQ2)
- ⑤ Limitations
- ⑥ Conclusion



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Amazon ranking objectives

Amazon pursues multiple objectives [4]:

- ▶ Relevance (R) [5, 1]
- ▶ Purchase likelihood (P) [5, 1]
- ▶ User engagement (E) [2]

Amazon ranking function

$$\text{Rank score} = f(x) = \max(\text{objective}(R, P, E)) \quad (1)$$

Amazon ranking features (x)

Sorokina and Cantú-Paz [5] reveal that Amazon uses the below ranking features:

Ranking features from Amazon Science

- ▶ **Behavioral features:** Clicks, views, add-to-cart, purchases.
- ▶ **Product features:** Sales data, reviews, popularity.
- ▶ **Query features:** Parsed attributes (e.g., brand, type) to infer intent.
- ▶ **Textual similarity:** Relevance between query and listing text (undisclosed metrics).

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Research gaps

- ▶ Most research focuses on **platform-side ranking models**, not seller strategies.
- ▶ Key ranking signals **remain private**; sellers lack guidance on which features of product listings to optimize.
- ▶ **Empirical studies on Amazon Search Engine Optimization (SEO) and product rank** are largely missing.

Research questions

To address the research gaps, this research investigates two questions to provide sellers actionable insights:

- ▶ **RQ1:** How does Amazon assign ranking to products on its search engine results pages (SERPs), and which **features contribute most significantly** to ranking?
- ▶ **RQ2:** What **factors help products maintain their rank** on Amazon SERPs over time?

Data collection

- ▶ **Step 1:** Selected 106 high-volume Amazon search queries.
- ▶ **Step 2:** Scraped SERP and product pages HTML content (pages 1–3, every 2 hours, for 7 days).
- ▶ **Step 3:** Parsed HTML to extract structured product features for analysis.

Final variables

Variable Name	Description
Relevance label	Graded score based on actual rank (1 is least relevant, 5 is most relevant).
Survival time	How long does the product maintain its rank before dropping to a to a lower rank
Discount	Whether a product's current price is lower than the original price (1), or not (0).
Initial rank	Initial position on the search results page.
Popularity	Whether a product's sales history, reviews, and sales rank are within the top 20%.
Price	The current price of a product.
Price competitiveness	Price rank of products within the same query group.
Prime	Whether a product is part of Amazon Prime (1), or not (0).
Rating	The average customer rating of a product.
Review count	The total number of reviews of a product.
Sales badge	Whether a product had sales in the past month above 50 (1), or not (0).
Sales rank main category	Sales position of the product relative to others in the main category.
Sales rank subcategory	Sales position of the product relative to others in the subcategory.
Sponsored	Indicates whether a product's position is due to paying (1), or not (0).
Semantic similarity	Semantic score between the product title, bullet points description and the query.
Lexical similarity	How often query terms appear in the product title, bullet points description (TF) and accounting for rareness of the term (TF-IDF).

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Methodology: Learning-to-rank (LTR) model

- ▶ **Goal:** approximate the Amazon ranking algorithm
- ▶ **Model:** XGBoost with rank:ndcg objective (LambdaMART algorithm)
- ▶ **Dataset:** 80-20% splitting, organic product listings only

Tabel: Input and output of model

Input (Product features)	Training target	Output
Discount	Relevance label	Relevance score
Popularity		
Price		
Price competitiveness		
Prime		
Rating		
Review count		
Sales badge		
Sales rank (main and sub category)		
Semantic similarity		
Lexical similarity (TF & IDF)		

Results: Model performance

- ▶ **Ranking quality metrics are sound** compared to the research in the field [3], especially considering using publicly scraped data.

@k	nDCG	Precision
Our model	0.859 (nDCG@10)	0.441 (avg)
Closest research [3]	0.8995 (nDCG@10)	0.7225 (Precision@3)

Results: Feature importance

- **Sales rank** is the most influential predictor of relevance, followed by **price competitiveness, semantic similarity, reviews, and ratings.**

Tabel: Top 10 Features

Feature	XGB Gain	XGB Imp.	SHAP	SHAP Imp.
Main category sales rank	0.169	Medium	0.457	High
Price competitiveness	0.030	Low	0.347	High
Semantic title	0.066	Medium	0.339	High
Sub-category sales rank	0.355	High	0.325	High
Log price	0.023	Low	0.195	Medium
Log review	0.061	Medium	0.186	Medium
Rating	0.042	Low	0.112	Medium
Semantic bullet	0.027	Low	0.090	Medium
Sales badges	0.044	Low	0.081	Medium
Discount	0.034	Low	0.081	Medium
Semantic description	0.018	Low	0.079	Medium

Implications for sellers

- ▶ **Sales performance** better sales ranks and sales badges increase relevance, especially within a product's category.
⇒ *focus on improving sales rank over absolute sales in the past month*
- ▶ **Sentimental features** like review count and rating moderately improve relevance.
⇒ *Amazon Vine or "Request a Review" to boost reviews and maintain ratings.*
- ▶ **Price competitiveness** is more important than absolute price or discount
⇒ *dynamic pricing to stay competitive and increase chances of being Featured Offer.*
- ▶ **Query-product similarity:** Semantic similarity is highly impactful, especially in titles.
⇒ *focus on meaningful, semantically rich content and avoid keyword stuffing.*

Outline

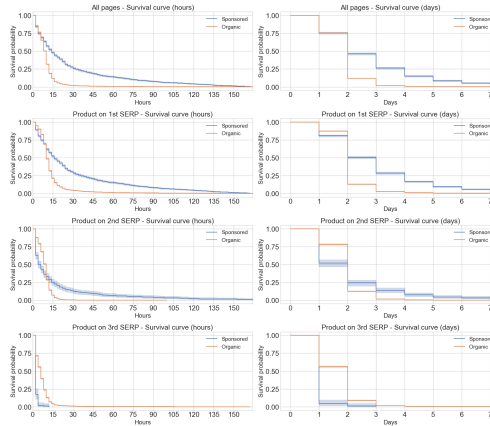
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Methodology: Survival analysis

- ▶ **Kaplan-Meier:** Plot probability curves to show survival trends over time.

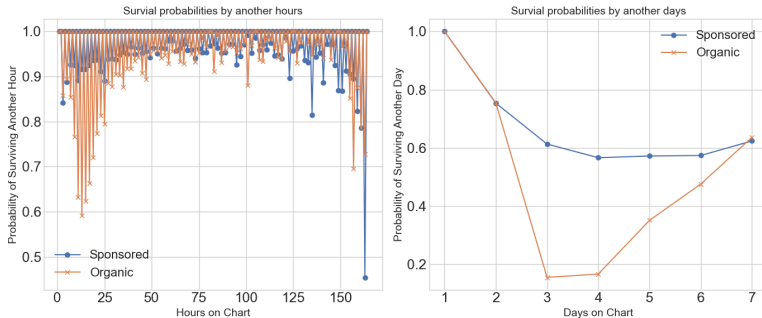
Results: Kaplan-Meier probability curves

- Distinct pattern and trend for sponsored vs. organic.



Results: Kaplan-Meier probability curves

- ▶ **Organic listings** drop off quickly, but those that survive tend to stabilize over time.
- ▶ **Sponsored listings** have higher survival probabilities but often decline gradually. Especially those on SERP 2–3, likely due to ad pacing strategies.



Methodology: Survival analysis

► Weibull Proportional Hazard Model (PHM)

- **Hazard ratio** reveals which features increase the risk of rank drop.

Tabel: Input and output of model

Input (Product features)	Output
Discount	Hazard ratio
Initial rank	
Popularity	
Price	
Price competitiveness	
Prime	
Rating	
Review count	
Sales badge	
Sales rank main category	
Sales rank subcategory	

Results: Weibull PHM results

- ▶ **Organic survival** is boosted by sales badges, and Prime.
- ▶ **Sponsored survival** is more influenced by pricing, especially for hourly (PC); fewer features are significant.

Feature	Organic model		Sponsored model	
	Hourly %	Daily %	Hourly %	Daily %
Prime	-5.5	-3.8		-13.0
Popularity	-30.3			
Sales badge	-33.1	-4.1		
Initial rank	+0.3	+0.2	+0.5	+0.1
Main rank	+5.9			+3.9
Sub rank	-2.5		-3.4	-2.1
Price	-2.0	-1.4		-4.2
Discount	-4.9			-5.9
Price competitiveness (PC)			-20.5	

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Limitations

- ▶ **Data:** no access to behavioral data (clicks, purchases, etc).
- ▶ **Algorithmic update:** findings may not fully reflect future ranking.
- ▶ **Generalization:** generalizing results to other e-commerce platforms is challenging.

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Conclusion

- ▶ Sellers should tailor their strategies based on their listing type and desired timeframe.
- ▶ Sellers should prioritize **sales and Prime**.
 - **SEO requires** optimizing additional features: review, discount, and popularity.
 - **SEA requires** optimizing price competitiveness, especially for short-term survival.

Feature	Rank Importance	Hourly Survival		Daily Survival	
		Organic	Sponsored	Organic	Sponsored
Sub-category sales rank	+++	↑	↑	○	↑
Main category sales rank	+++	↓	○	○	↓
Sales badge	++	↑	○	↑	○
Price competitiveness	+++	○	↑	○	○
Semantic title	+++				
Review	++	↑	○	○	○
Rating	++	↓	○	○	○
Semantic bullet	++				
Price	++	↑	○	↑	↑
Discount	++	↑	○	○	○
Semantic description	++				
Prime	+	↑	○	↑	↑
Popularity	+	↑	○	○	○
Initial rank		↓	↓	↓	↓

Note: Ranking importance: +++ = High, ++ = Medium, + = Low. Survival Impact: ↑ = Improves survival (reduces rank drop risk), ↓ = Reduces survival (increases rank drop risk), ○ = Not significant.

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Appendix: Ranking model (Answer RQ1)

- **Hyperparameter tuning:** random search is used to find the best parameters

Parameters	Value
Learning rate	0.19
Tree depth	5
Number of trees	161
Sample rate	55%
Feature sample	83%
Min sample split	2
Compared pairs	16
Pair method	Mean

Appendix: Ranking model (Answer RQ1)

► Evaluation Metrics (eg.)

- **Precision@5** = 0.60 (3/5 match: A, B, C)
- **nDCG@5** $\approx$ **0.85** (A is ranked 1, but penalized for F,G at positions 2,3)

Rank	Model prediction	Actual top 4 Amazon
1	A (relevance label=5)	A (relevance label=5)
2	F (relevance label=1)	B (relevance label=4)
3	G (relevance label=1)	C (relevance label=3)
4	B (relevance label=4)	D (relevance label=2)
5	C (relevance label=3)	H (relevance label=1)

► Feature Importance:

- **XGBoost Gain**: Shows which features help the model most.
- **SHAP**: Explains how each feature affects each product's rank.

Appendix: Survival analysis (Answer RQ2)

- ▶ **Goal:** Analyze how long products stay in top search ranks (survival time).
- ▶ **Design:**
 - Survival time = the longest streak a product stays at/above its initial rank.
 - Event = when the product drops below that rank.
 - Compare organic vs sponsored listings for SEO vs SEA insights.

Datetime	Rank	Rank change	Streak ID	Description
08:00	5	–	0	Start (Streak 1)
10:00	4	-1	0	Survived
12:00	6	+2	1	Dropped – (Streak 2)
14:00	5	-1	1	Survived
16:00	3	-2	1	Survived
18:00	7	+4	2	Dropped – (Streak 3)
20:00	6	-1	2	Survived

Appendix: Query

- Queries are mostly Electronics and Home & Kitchen categories

Category	# Keywords
All Departments	5
Baby Products	1
Beauty	3
Electronics	23
Fashion	9
Garden	1
Grocery	2
Health & Personal Care	16
Home & Kitchen	25
Office Products	2
Pet Supplies	4
Sports & Outdoors	4
Tools & Home Improvement	3
Toys	1
Video Games	7

SERPs



- Sales badge

Product pages



Bullet points

Appendix: SHAP summary

